AI – Powered Stock Market Insight Generator for the Colombo Stock Exchange

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Project Proposal Report

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Component: Financial Document Understanding and Language-Driven Insight Extraction

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Co- Supervisor Name	Date	Signature
Dr. Mahima Weeerasinghe		

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ABSTRACT

Decisions about stock market investments require careful interpretation of financial information published by companies. In the context of the Colombo Stock Exchange (CSE), listed companies regularly publish financial reports containing critical indicators such as revenue, profitability, debt levels and growth trends. However, these reports are often long, complex and filled with domain-specific financial terminology, making them difficult for retail investors to interpret. As a result, many investors fail to extract meaningful insights and instead rely on speculation or incomplete information, leading to suboptimal investment decisions.

Although there are several global financial analysis platforms, they primarily focus on numerical stock data, technical indicators and price visualizations rather than interpreting unstructured financial documents. Furthermore, there is a significant lack of intelligent systems tailored to the Sri Lankan market that can analyze company financial disclosures and transform them into simplified, explainable insights for investors. This highlights a clear research gap in the development of a system capable of understanding financial reports and providing context-aware, investor-friendly interpretations in an emerging market.

To solve this problem, this research proposes a retrieval-enhanced AI-powered financial document reasoning system that goes beyond traditional summarization. The system is designed to automatically extract key financial indicators, interpret narrative disclosures and generate structured, explainable investor insights. It incorporates Natural Language Processing (NLP) techniques for financial entity extraction and integrates Retrieval-Augmented Generation (RAG) to incorporate relevant contextual knowledge such as financial concepts, historical data patterns and domain-specific references. This enables the system to produce more accurate, grounded and context-aware explanations of company performance.

The proposed solution will be implemented using Python, advanced NLP techniques, document processing tools and large language models (LLM), enhanced with a retrieval mechanism to support context-aware reasoning. The system will transform unstructured financial reports into structured investor signals, including financial health indicators, risk assessments and growth insights, presented in a simplified and understandable manner.

The expected result is an intelligent and explainable financial insight generation module that improves financial literacy, supports data-driven investment decisions and reduces reliance on speculative behavior among retail investors. This research also offers both social and commercial value by introducing a scalable AI-powered solution tailored for the Sri Lankan stock market.

Keywords: Artificial Intelligence, Financial Document Analysis, Retrieval-Augmented Generation (RAG), Natural Language Processing, Large Language Models, Explainable AI, Stock Market Insights.

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LIST OF ABBREVIATIONS

Table 1:List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSE	Colombo Stock Exchange
CSV	Comma Separated Values
DL	Deep Learning
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PDF	Portable Document Format
SaaS	Software as a Service
UI	User Interface

1. Introduction

1.1 Background and Context

Stock markets play a critical role in economic development by enabling companies to raise capital and providing investment opportunities for individuals. The Colombo Stock Exchange (CSE), the primary stock exchange in Sri Lanka, comprises over 290 listed companies across multiple sectors, including banking, manufacturing and services. Investors rely heavily on financial information such as annual reports, quarterly reports and regulatory filings to evaluate company performance and make informed investment decisions.

However, interpreting financial reports remains a significant challenge, especially for private investors who often lack advanced financial and analytical expertise. These reports are typically long, complex, and contain domain-specific financial terminology along with extensive unstructured textual information. Extracting meaningful insights such as profitability trends, risk factors and financial stability requires specialized knowledge. Consequently, many investors struggle to accurately assess a company's financial health and may rely on speculation or incomplete information, increasing the likelihood of poor investment decisions and financial risk.

Globally, the participation of private investors has increased significantly in recent years. Despite this growth, a large proportion of investors rely on simplified summaries, third-party opinions or market rumors rather than directly analyzing financial disclosures. In Sri Lanka, the growing engagement of retail investors on the Colombo Stock Exchange further underscores the need for intelligent, accessible tools that can simplify financial information and support data-driven decision-making.

Existing financial platforms such as Bloomberg Terminal, Yahoo Finance and TradingView provide extensive numerical market data, price visualizations and analytical tools. However, these platforms primarily focus on structured data and do not interpret complex financial documents efficiently. In addition, many advanced financial analysis tools are expensive and designed for professional analysts, limiting their availability to ordinary investors. As a result, there is a clear gap in systems capable of making sense of unstructured financial reports and generating simplified, explainable insights tailored for retail investors, especially in emerging markets such as Sri Lanka.

To address this limitation, this research proposes the development of a retrieval-enhanced AI-powered financial document understanding and reasoning module. Unlike traditional approaches that rely solely on summarization, the proposed system is designed to extract key financial indicators, interpret narrative disclosures, and generate structured, explainable investor insights. The system incorporates Natural Language Processing (NLP) techniques for extracting financial information and integrates Retrieval-Augmented Generation (RAG) to improve contextual understanding by retrieving relevant financial knowledge, historical references and domain-specific information. This makes it possible to generate more accurate, grounded and context-aware explanations of company performance.

The proposed module transforms unstructured financial data into structured investor signals such as financial health indicators, risk assessments and growth insights. These outputs can be integrated with other system components, including market trend analysis, sentiment analysis and investor dashboards, forming a comprehensive decision support system.

This research is in line with Sustainable Development Goal (SDG) 8 – Decent work and economic growth by promoting financial literacy and enabling informed participation in financial markets. Furthermore, it contributes to the research domain of Artificial Intelligence and Data Analytics by promoting the application of retrieval-extended large language models for understanding financial documents and explainable decision support systems in emerging market contexts.

1.2 Literature Review

Recent advances in artificial intelligence (AI) and natural language processing (NLP) have significantly improved the ability of machines to analyze and interpret complex textual data. In the financial field, researchers have increasingly explored the use of AI techniques to extract meaningful insights from financial documents such as annual reports, income statements and regulatory disclosures. These documents contain valuable information about the company's performance, financial risks and strategic decisions, which are important to investors and analysts.

Several studies have focused on using Natural Language Processing techniques for financial text analysis. For example, Loughran and McDonald highlighted that financial documents contain specialized vocabulary and domain-specific language that differ significantly from general text corpora. Their research showed that financial sentiment analysis requires custom dictionaries and models tailored specifically for financial documents. Similarly, Kogan et al. investigated the use of text mining techniques to analyze financial disclosures and found that textual information in annual reports can provide valuable predictive insights into corporate performance.[1][2]

With the emergence of large language models (LLM) such as GPT-based architectures, researchers have begun to explore their application in financial document understanding and summarization. Brown et al. demonstrated that large-scale language models can perform various language comprehension tasks, including summarization and information extraction, with minimal task-specific training. Recent studies have investigated how LLMs can be used to generate simplified explanations from complex financial reports to help non-expert users understand financial data.[3][4]

In addition to academic research, several commercial platforms offer analysis tools for financial data. Systems such as Bloomberg Terminal, Yahoo Finance and Trading View offer comprehensive financial datasets, stock market indicators and analytical tools for investors. However, these platforms mainly focus on numerical market data and financial charts rather than automatic interpretation of financial documents. Also, professional platforms like Bloomberg Terminal are often expensive and designed primarily for institutional investors rather than retail investors.[5]

Recent studies have also explored automated summarization of financial reports using machine learning techniques. El-Haj et al. developed methods for analyzing company annual reports and extracting key information from narrative parts of financial documents. Similarly, Huang et al. proposed transformer-based approaches for financial text analysis that improve the extraction of financial indicators and contextual meaning from financial documents.[6][7]

Despite these advances, existing research and systems still face several limitations. Many current studies focus on general financial text datasets rather than real financial documents from emerging markets. In addition, most existing platforms do not convert complex financial disclosures into simplified insights that can assist retail investors in their decision-making process. There is also limited

research that focuses on localized financial environments such as the Colombo Stock Exchange, where financial data structures, reporting formats and investor needs may differ from global markets.

Therefore, there is a need for a system that integrates financial document processing, Natural Language Processing and Large Language Models to automatically extract key financial indicators and generate simplified investor-friendly insights from the company's financial reports. The proposed research aims to address this gap by developing an AI-based financial document understanding module tailored to the Sri Lankan stock market environment.

1.3 Research Gap

Although recent advances in artificial intelligence (AI) and natural language processing (NLP) have greatly improved the ability to analyze text data, several limitations still exist when applying these techniques to financial document analysis. Existing financial platforms such as Bloomberg Terminal, Yahoo Finance and TradingView focus primarily on structured market data, including stock prices, technical indicators and economic trends. While these systems provide valuable numerical insights, they do not automatically interpret complex financial documents such as annual reports, income statements and regulatory disclosures. Consequently, investors are required to manually analyze long and complex reports to extract meaningful insights, which is particularly challenging for retail investors with limited financial expertise.

Current research in financial text analytics has demonstrated the effectiveness of machine learning and NLP techniques for extracting information from financial data. However, a majority of these studies focus on sentiment analysis of financial news or short text data rather than deep understanding of comprehensive financial reports. Moreover, many existing approaches have been developed using global datasets and do not take sufficient account of the unique characteristics of emerging markets such as Sri Lanka, where reporting structures, financial terminology and investor needs are significantly different.

Another critical limitation is that traditional NLP and rule-based systems lack the ability to capture contextual and domain-specific meaning within complex financial disclosures. Although large language models (LLMs) have shown strong capabilities in language understanding and summarization, their stand-alone application often lacks grounding, leading to potential inaccuracies or hallucinatory output. Furthermore, existing systems rarely integrate contextual financial knowledge or historical references when interpreting financial reports, limiting their ability to generate reliable and explainable insights.

Therefore, there is a clear research gap in developing a system that not only extracts financial information from unstructured documents, but also contextualizes, interprets and transforms this information into structured, explainable investor insights. In particular, there is a lack of systems that combine document understanding, contextual knowledge acquisition and economic reasoning tailored to emerging market environments.

To address this gap, this research proposes a retrieval-enhanced financial document understanding and reasoning module that integrates NLP-based financial information extraction with retrieval augmented generation (RAG). By incorporating relevant contextual knowledge such as economic concepts, historical patterns and domain-specific references, the system improves the accuracy, reliability and explainability of generated insights. Unlike traditional approaches that focus on summarization, the

proposed system transforms unstructured financial disclosures into structured investor signals, including financial health indicators, risk assessments, and growth insights.

The novelty of this research lies in the development of a localized, retrieval-augmented economic reasoning framework for the Colombo Stock Exchange, which combines the understanding of financial documents, the retrieval of contextual knowledge and the generation of explainable insights. This approach addresses key limitations in existing systems by enabling context-aware, anchored and investor-oriented analyzes of financial reports within an emerging market setting. Ultimately, this helps improve financial access, reduce information asymmetry and support more informed investment decisions.

2.OBJECTIVE

2.1 Main Objective

To design and implement a retrieval-augmented AI-based financial document understanding and reasoning system that extracts key financial indicators, incorporates contextual knowledge, and generates structured, explainable investor insights from Colombo Stock Exchange company reports using large language models, achieving high accuracy, contextual relevance, and reliability.

2.2 Specific Objectives

- To collect and preprocess a dataset of at least 100 financial documents, including annual reports, income statements, and regulatory disclosures from companies listed on the Colombo Stock Exchange.
- To design and implement a document processing pipeline capable of extracting both textual and numerical financial data from PDF reports with a minimum of 85% text extraction accuracy.
- To develop a financial information extraction module that identifies key financial indicators, financial ratios, and risk-related factors from financial documents with a minimum precision of 80%.
- To design and implement a retrieval mechanism (RAG) that retrieves relevant contextual financial knowledge, including domain concepts, historical references, and related financial data to support document interpretation.
- To integrate large language models (LLMs) with the retrieval mechanism to perform context-aware financial reasoning and generate accurate, grounded, and investor-friendly explanations.
- To transform unstructured financial data into structured investor signals, including financial health indicators, risk assessments, and growth insights, suitable for downstream analytical modules and dashboards.
- To benchmark and evaluate different model configurations based on extraction accuracy, contextual relevance, explainability, and reliability of generated insights.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) approach, which focuses on the design, development and evaluation of an artifact to solve a real-world problem. The artifact developed in this study is a retrieval-extended AI-powered financial document understanding and reasoning module, designed to extract, interpret and generate structured investor insights from financial reports of companies listed on the Colombo Stock Exchange (CSE). The DSR approach is appropriate as the research aims to create and evaluate a practical system that improves the interpretation of financial documents and supports informed investment decisions.

3.1 Technical methods and algorithms

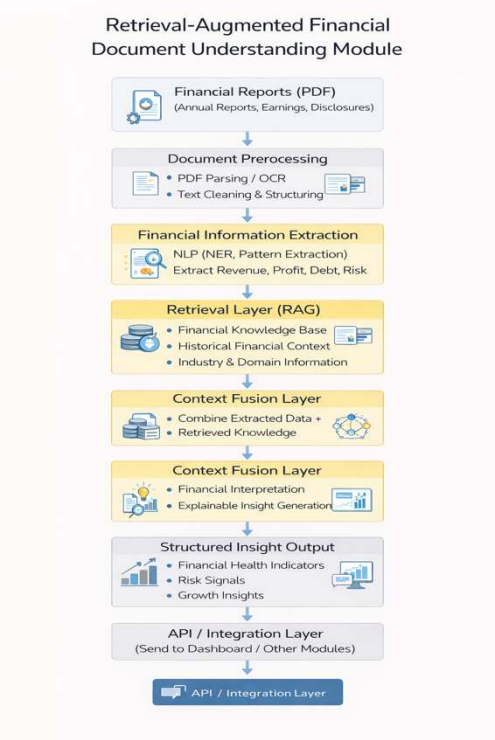


Figure 1: AI Processing Pipeline Diagram

The proposed system follows a multi-stage processing pipeline to analyze financial documents and generate structured investor insights.

Firstly, financial reports such as annual reports, income statements and regulatory information will be collected from publicly available sources relating to companies listed on the Colombo Stock Exchange (CSE). These documents are usually available in PDF format and contain both structured and unstructured financial information.

Next, document preprocessing and parsing techniques will be used to extract textual content from PDF files. This includes using PDF analysis tools and OCR (Optical Character Recognition) techniques where necessary to convert document content into machine-readable text. The extracted text will be cleaned and structured to improve processing quality.

After pre-processing, Natural Language Processing (NLP) techniques will be used to identify and extract important financial information. Information mining methods such as Named Entity Recognition

(NER) and pattern-based approaches will be used to discover financial indicators including revenues, profit margins, debt ratios and risk-related factors from the reports.

To improve contextual understanding and overcome limitations of stand-alone language models, the system integrates a RAG (Retrieval-Augmented Generation) mechanism. This component retrieves relevant contextual knowledge such as financial concepts, historical financial data and industry-specific information, which is then combined with the extracted document data.

Large language models (LLMs) are then used as a reasoning engine to interpret both the retrieved financial data and the retrieved contextual information. Rather than simple summarization, the system generates structured, explainable investor insights, including financial health indicators, risk signals and growth-related insights. Rapid engineering techniques will be used to guide the model in producing clear, accurate and investor-friendly results.

Finally, the generated insights are transformed into structured results that can be integrated with other system components such as market analysis, sentiment analysis and investor dashboards.

To ensure reliability and performance, several LLM configurations and retrieval strategies will be benchmarked and evaluated based on metrics such as information extraction accuracy, contextual relevance, explainability, and consistency of generated insights.

3.2 Tools and platforms

The proposed system will be implemented using a combination of modern Artificial Intelligence, language Processing, and data engineering tools.

Python will be used as the primary programming language due to its extensive ecosystem for machine learning, natural language processing and data analytics. NLP libraries such as spaCy, NLTK and Hugging Face Transformers will be used for text preprocessing, tokenization and financial information extraction. Document parsing will be performed using tools such as PyPDF, PDFMiner and OCR-based solutions where necessary, enabling the extraction of both structured and unstructured content from financial reports.

To support contextual understanding, the system will include a Retrieval-Augmented Generation (RAG) framework, which integrates a retrieval layer with large language models. This will be implemented using embedding models and vector databases (eg FAISS or similar tools) to retrieve relevant financial knowledge, historical context and domain-specific information.

Large language models (LLM), accessible via platforms such as OpenAI APIs or transformer-based models from Hugging Face, will be used as a reasoning engine to interpret mined financial data and generate structured, explainable investor insights. Rapid engineering strategies will be used to ensure that the output is accurate, context-aware and adapted to economic interpretation requirements.

Data processing and transformation will be handled using libraries such as Pandas and NumPy, while integration with other system components will be supported through API-based communication.

These tools have been selected for their strong support for scalable AI development, suitability for financial text analytics, and ability to support retrieval-augmented, context-aware reasoning systems.

3.3 Data collection and ethical considerations

The data used in this research will consist of publicly available financial reports and regulatory disclosures published by companies listed on the Colombo Stock Exchange (CSE). As these documents are officially released for investor communications, their use for research purposes does not violate privacy or data protection regulations.

However, ethical considerations will be strictly observed throughout the research process. The system is designed as a decision support tool, not a financial advisory system, and will not provide buy or sell recommendations.

To ensure responsible AI use, the system will:

- Generate accurate and grounded insights based on mined financial data and derived contextual knowledge
- Avoid hallucinating or misleading output using retrieval-enhanced mechanisms (RAG)
- Provide explainable output that reflects the original financial disclosures
- Maintain transparency by ensuring that insights generated can be traced back to source information

In addition, the system will aim to minimize bias and misinterpretation by validating output across multiple model configurations and ensuring consistency in financial interpretation.

Overall, the research promotes ethical AI use by improving financial literacy while maintaining accuracy, transparency and accountability in generated insights

.3.4 Validation strategy and performance measurements

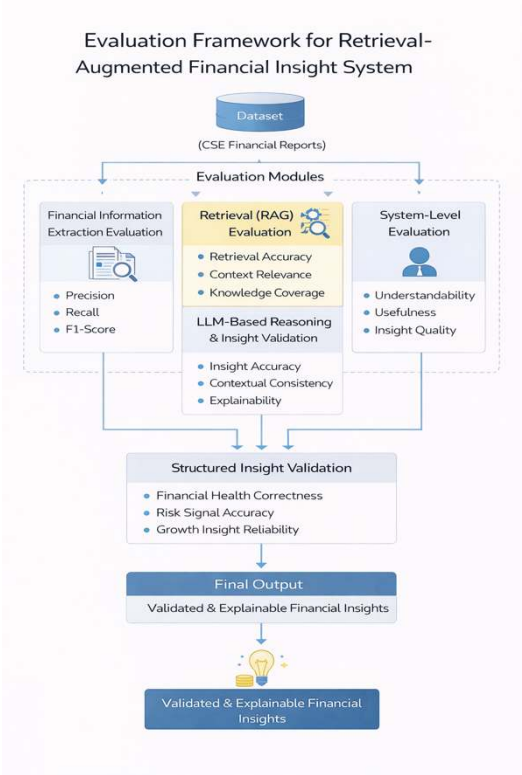


Figure 2: Validation Process Diagram

To evaluate the effectiveness of the proposed system, both quantitative and qualitative performance metrics will be used across various components of the pipeline.

The financial information extraction module will be evaluated using standard information retrieval metrics including Precision, Recall and F1 scores. These metrics will measure the accuracy and completeness of extracting key financial indicators, ratios and relevant textual information from the company's financial reports and regulatory disclosures.

The large language model (LLM)-based summary component will be evaluated through qualitative and comparative analysis. The generated summaries will be assessed based on their relevance, contextual correctness and completeness in representing the original financial information. In addition, several LLM configurations will be benchmarked to identify the most effective model for interpreting financial documents within the context of the Colombo Stock Exchange.

Furthermore, the overall system performance will be evaluated using a qualitative assessment of the generated financial insight. This evaluation will focus on the clarity, interpretability and usefulness of the explanations provided to retail investors. The aim is to ensure that the system produces meaningful and understandable insights that effectively support financial literacy and informed decision-making.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

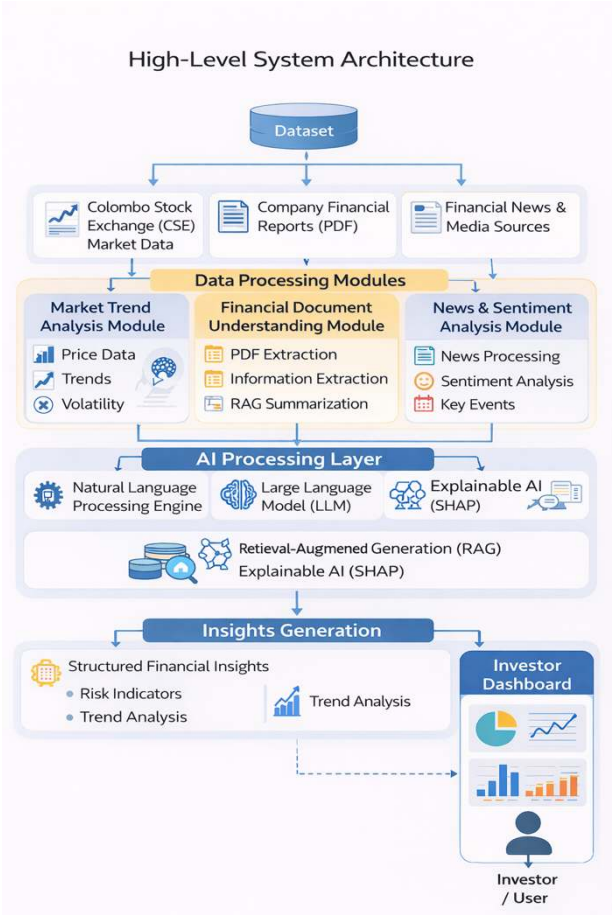


Figure 3:High-Level System Architecture of the Proposed AI-Powered Stock Market Insight System

The proposed solution is an AI-powered Stock Market Insight System designed for the Colombo Stock Exchange (CSE). The main objective of the system is to support retail investors by providing explainable, data-driven insights rather than direct buy or sell recommendations. By focusing on interpretability and transparency, the system helps users understand the reasoning behind market behavior. It integrates multiple analytical components to process various financial data sources, including historical market data, company financial reports and financial news articles. These components collaboratively generate structured insights, risk indicators and contextual explanations, which are presented through a unified investor dashboard.

The system uses three main data sources: Colombo Stock Exchange market data, corporate financial reports in PDF format and financial news and media content. Each data source is handled by a dedicated analysis module. The Market Trend Analysis Module processes historical price and trading volume data to identify trends, volatility and market patterns. The Financial Document Understanding Module, which is the main focus of this research component, analyzes annual reports, profit and loss accounts and regulatory disclosures to extract key financial indicators, key financial figures and risk-related information. In parallel, the News and Sentiment Analysis Module processes financial news and event-based content to detect sentiment signals and identify external factors that influence stock market behavior.

The individual contribution to this research is the Financial Document Understanding and Language-Driven Insight Extraction Module. This module consists of several internal stages, including document collection, text mining, preprocessing, financial information extraction, large-language model-based summarization, and structured insight generation. Through these stages, complex financial documents are transformed into simplified, investor-friendly explanations. The output from this module is converted into structured financial insights, which can be used by other analytical components and visualized through the system dashboard. This component plays a critical role in improving financial literacy by enabling investors to interpret complex financial information in an accessible and meaningful way.

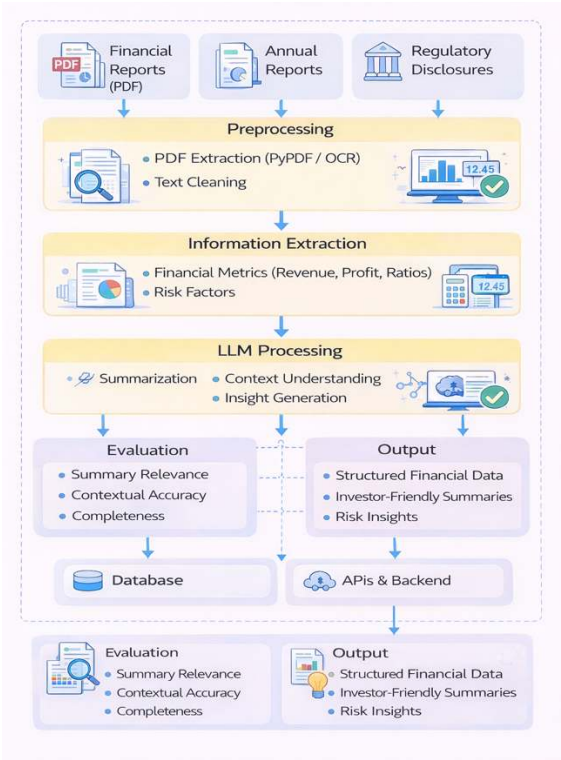


Figure 4:Architecture of the Financial Document Understanding and Language-Driven Insight Extraction Module

The interaction between system components follows a structured layered data flow architecture. Data obtained from market sources, financial documents and news sources are first processed within their respective analytical modules. The output from these modules is then forwarded to the Natural Language Processing (NLP) Engine and the Large Language Model (LLM) processing layer. At this stage, the extracted information is refined, interpreted and transformed into meaningful, actionable insights. The final results are generated as structured financial insights and risk indicators, which are then integrated into the Investor Dashboard. Through this dashboard, users can interact with the system to explore explanations, market trends and financial indicators in an intuitive and user-friendly way. This architecture enables efficient integration of structured, semi-structured and unstructured data into a unified and explainable insight generation pipeline.

From a scalability perspective, the proposed architecture is modular in design, allowing individual components to be independently enhanced or extended without requiring a complete system redesign. This flexibility enables the seamless integration of additional data sources, advanced AI models, or new dashboard functionalities as the system evolves. From a security and reliability standpoint, the system primarily utilizes publicly available financial data, thereby minimizing privacy-related concerns. However, to ensure the quality and trustworthiness of the generated insights, validation mechanisms and controlled model outputs are incorporated to mitigate the risk of inaccurate or misleading AI-generated summaries. This approach ensures that the system remains reliable, interpretable, and suitable for supporting informed investment decisions.

5.USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-powered stock market insight system involves multiple stakeholders who interact with or benefit from the platform.

Retail Investors:

Retail investors are the primary users of the system. Many of them lack the expertise required to interpret complex financial reports and market data. The system assists them by providing simplified, explainable insights related to financial performance and risk indicators, enabling more informed decision-making.

Financial Analysts:

Financial analysts can utilize the system to efficiently analyze company financial reports, identify key financial indicators, and observe emerging trends without manual data processing.

Brokerage Firms:

Brokerage firms can integrate the system into their services to support clients by offering AI-generated insights and risk indicators derived from financial data, thereby enhancing advisory capabilities.

Researchers and Academic Users:

Researchers and academic users can leverage the system to study financial patterns, sentiment analysis techniques, and AI-based financial document understanding methodologies, contributing to further research and development in this domain.

5.2 Functional Requirements

The system must provide the following functional capabilities:

1. The system shall allow users to upload company financial reports in PDF format.
2. The system will extract text content and financial data from uploaded financial reports.
3. The system must identify key financial indicators such as income, profit, debt ratio and financial risk factors.
4. The system will use natural language processing techniques to analyze financial documents.
5. The system will integrate large language models to generate simplified summaries of financial reports.
6. The system must convert unstructured financial data into structured financial insight.
7. The system shall generate financial risk indicators based on extracted financial information.
8. The system shall display generated insight and risk indicators through an investor dashboard.

5.3 Non-Functional Requirements

Security:
The system shall ensure that uploaded financial documents and generated insights are securely processed and stored, preventing unauthorized access.

Performance:
The system shall process uploaded financial reports and generate insights within an acceptable response time to ensure usability.

Scalability:
The system architecture shall support future expansion, including the integration of additional data sources, analytical modules, and advanced AI models.

Usability(Ease Of Use):

The system interface shall present financial insights in a clear, intuitive, and user-friendly manner, suitable for non-expert users.

Reliability:
The system shall consistently produce accurate and dependable outputs when processing financial documents.

Maintainability:
The modular architecture shall enable easy updates, debugging, and enhancement of individual components such as the NLP engine and LLM processing modules.

6.COMMERCIALIZATION PLAN

The proposed AI-powered stock market insight system shows strong commercialization potential in the fintech sector, especially for stakeholders operating on the Colombo Stock Exchange (CSE). The primary target market includes retail investors, financial analysts and brokerage firms who need effective tools to interpret complex financial reports and market data. The system delivers value by automatically analyzing the company's accounts, market trends and financial news to generate simplified, investor-friendly insights and financial risk indicators.

A Software-as-a-Service (SaaS) business model is proposed for commercialization. Under this model, core features such as basic report analysis and limited insights can be offered for free, while advanced features – including detailed financial analysis, risk modeling and enhanced insights – are offered through subscription-based premium plans. This freemium pricing strategy enables user acquisition at scale while generating sustainable revenue from power users and institutional customers.

The primary operating costs of the system include cloud infrastructure, deployment and maintenance of AI models, data processing, and continuous system updates. Scalable cloud-based solutions can be used to manage compute requirements efficiently while minimizing operational costs.

The main competitive advantage of the proposed system lies in its ability to integrate financial document understanding, market trend analysis and sentiment analysis into a unified platform using large language models tailored specifically for the Sri Lankan stock market context. Unlike existing global platforms, this system emphasizes localized financial interpretation and explainable insights, making it more relevant and accessible to local investors.

From an intellectual property perspective, the system includes several valuable assets such as the overall software architecture, financial document processing pipeline, and AI-driven analysis methodologies developed during this research. These components can be protected through software copyright and may also be considered for future patent applications, depending on the level of innovation and technical implementation.

7.BUDGET AND JUSTIFICATION

The development of the proposed AI-powered stock market insight system requires multiple resources, including hardware, software tools, cloud infrastructure, and system development. The estimated budget for the project is presented below.

Table 2: Budget Table

Category	**Item**	**Estimated Cost (LKR)**	**Justification**
Hardware	Personal laptop / workstation	0 (existing)	The research will be conducted using the student's existing laptop capable of running development tools and machine learning frameworks.
Software Licenses	Development tools (Python, NLP libraries)	0	Most development tools such as Python, TensorFlow, PyTorch, and NLP libraries are open-source and freely available.
Cloud Services	Cloud computing resources	LKR 30,000	Cloud services may be required for running large language models, storing datasets, and processing financial documents.
API Services	LLM API usage (OpenAI or similar)	LKR 40,000	API access may be required to test and benchmark multiple large language model configurations for financial document summarization.
Data Storage	Cloud database / storage services	LKR 10,000	Required for storing financial reports, extracted data, and generated insights during experimentation.
Human Effort	Research and development effort	Not monetized	The system development and experimentation will be conducted as part

			of the academic research project.

Total estimated cost: Approximately LKR 80,000

Most of the development tools used in this research are open-source technologies, which significantly reduce software license costs. The main expenses are related to cloud computing services and API usage required for processing financial documents and evaluating large language models. Since the research is carried out as part of a bachelor's thesis, the cost of human effort is not monetized, but represents the time and technical effort invested in developing the system.

8.WORK BREAKDOWN STRUCTURE (WBS)

The work breakdown structure defines the main tasks required for the development of the AI-powered stock market insight system. Tasks are divided between team members according to their research components to ensure balanced workload distribution and effective project management. The timeline is in line with the research project's milestones from January to November.

Table 3:Work Breakdown Structure Table

Phase	Task	Responsible Member	Timeline
Project Planning	Literature review and research gap identification	All Members	Jan – Feb
System Design	High-level architecture and system design	All Members	Feb
Data Collection	Collect market data, financial reports, and news data	Samarathunga, Fernando, Harithra	Feb – Mar
Market Analysis Module	Develop ML models for market trend and risk analysis	Samarathunga	Mar – May
Financial Document Processing	PDF parsing, OCR, and text preprocessing	Fernando	Mar – May
Financial Information Extraction	Extract financial indicators, ratios, and risk factors	Fernando	Apr-May
LLM Insight Generation	Implement LLM-based financial summary generation	Fernando	Apr – Jun
Sentiment Analysis	Develop NLP-based news sentiment analysis	Harithra	Apr – Jun
Dashboard Development	Implement investor dashboard and visualizations	Wijesooriya	May – Jul
System Integration	Integrate all modules and APIs	All Members	Jul – Aug
System Testing	Perform performance evaluation and validation	All Members	Aug – Sep
Optimization	Improve model accuracy and usability	All Members	Sep
Documentation	Prepare final report and research paper	All Members	Sep – Oct
Final Deployment	Deploy system and prepare demonstrations	All Members	Oct – Nov

The work breakdown structure divides the project tasks among team members based on their specialized research components. Each member is responsible for developing and evaluating a specific module of the system, including market analysis, financial document understanding, sentiment analysis and visualization. The tasks are planned across the project timeline from January to November, which ensures a balanced distribution of workload and effective coordination between team members. Regular integration and testing phases ensure that all modules work together as a unified AI-powered insight generation system.

9.GANTT CHART

Project Activity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Topic selection & supervisor discussions	■										
Proposal preparation	■	■	■								
Proposal submission & presentation			■								
Literature review & system design		■	■	■							
Data collection (CSE data, reports, news)			■	■	■						
Core module development				■	■	■	■				
Progress Presentation 1					■						
System integration & dashboard development					■	■	■	■			
Progress Presentation 2								■			
Testing & performance evaluation							■	■	■		
Final report writing								■	■	■	
Final report submission										■	
Website submission										■	
Final presentation & viva										■	
Research paper submission										■	
Research paper publication evidence											■

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